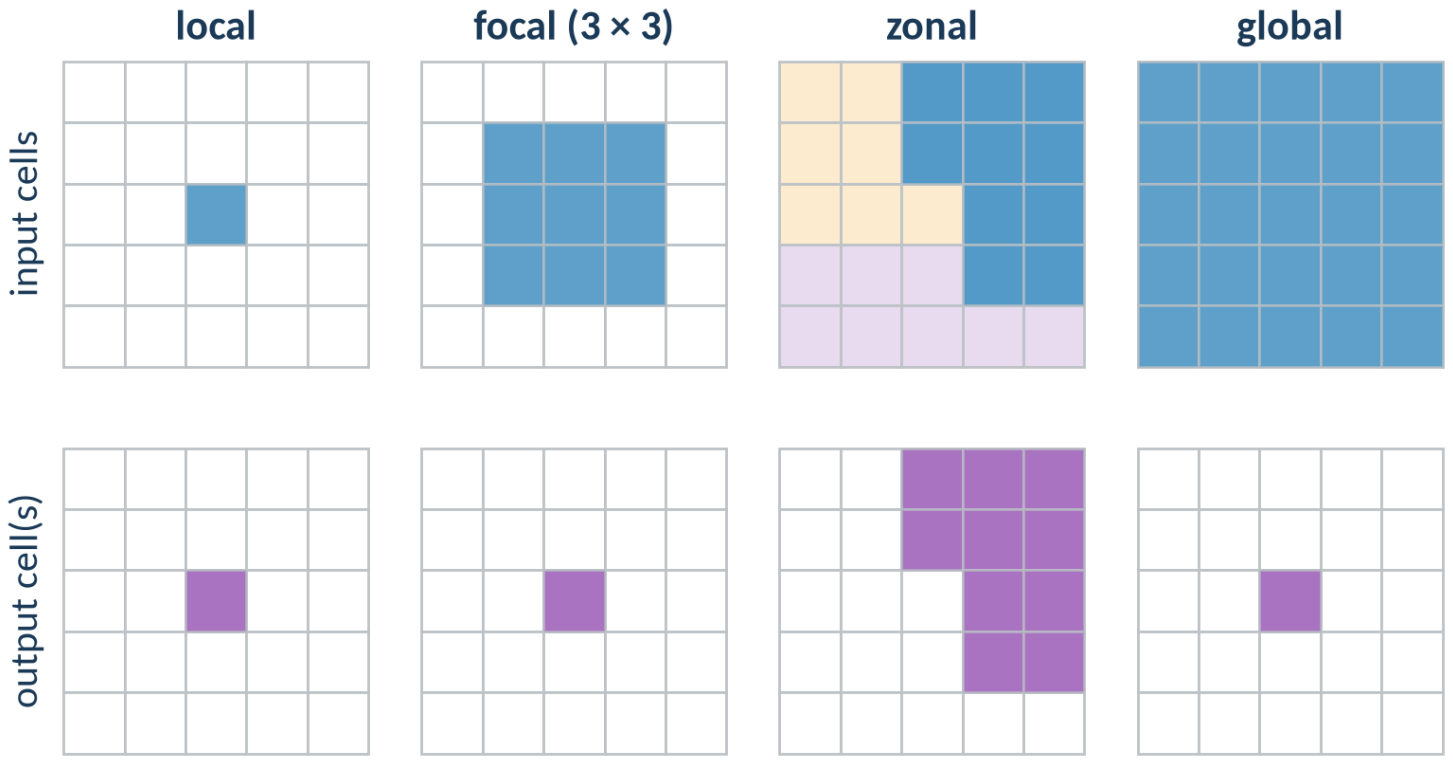
Raster operations

Map algebra: local, focal, zonal and global operations on aligned grids

Examples: Sentinel-2 and Copernicus DEM, Kamp-Lintfort



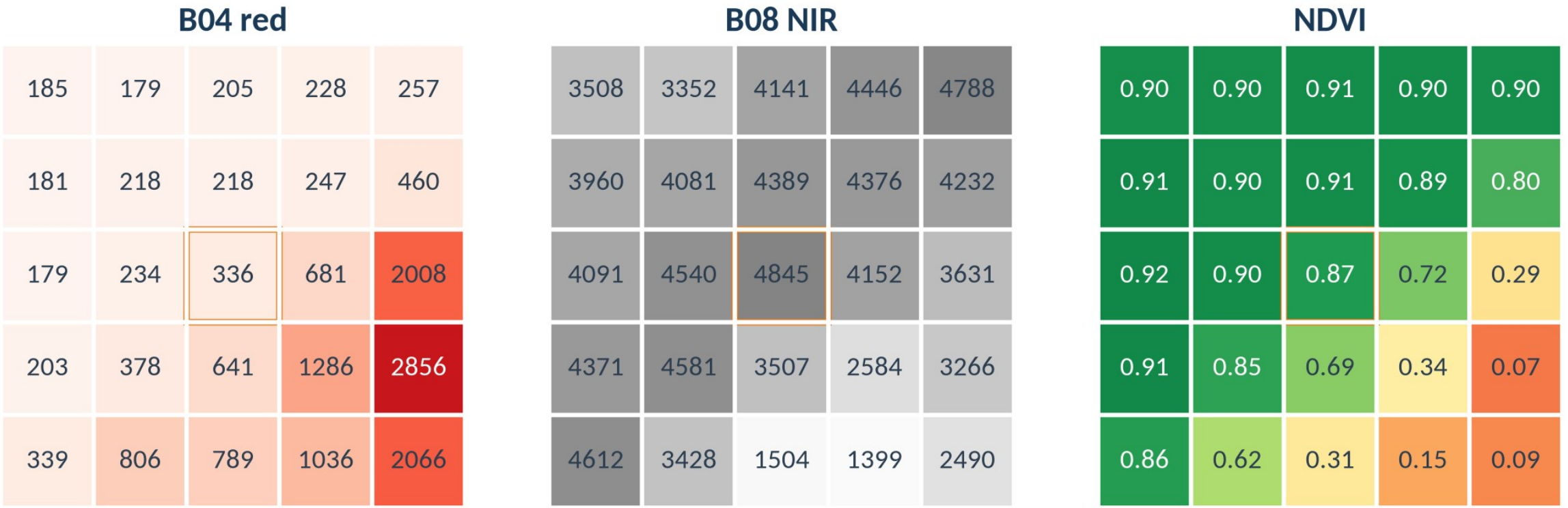
Map algebra: which cells feed the result?



- **Local:** cell by cell (NDVI, A + B)
- **Focal:** moving window (slope, smoothing)
- **Zonal:** per zone (mean NDVI per land cover)
- **Global:** whole grid (distance, min/max)

After C. D. Tomlin (1990): Geographic Information Systems and Cartographic Modeling

Local operation, cell by cell: NDVI



orange cell: (4845 − 336) / (4845 + 336) = 0.87

```
red, nir = s2[2], s2[3]  # float32
ndvi = (nir - red) / (nir + red)
```

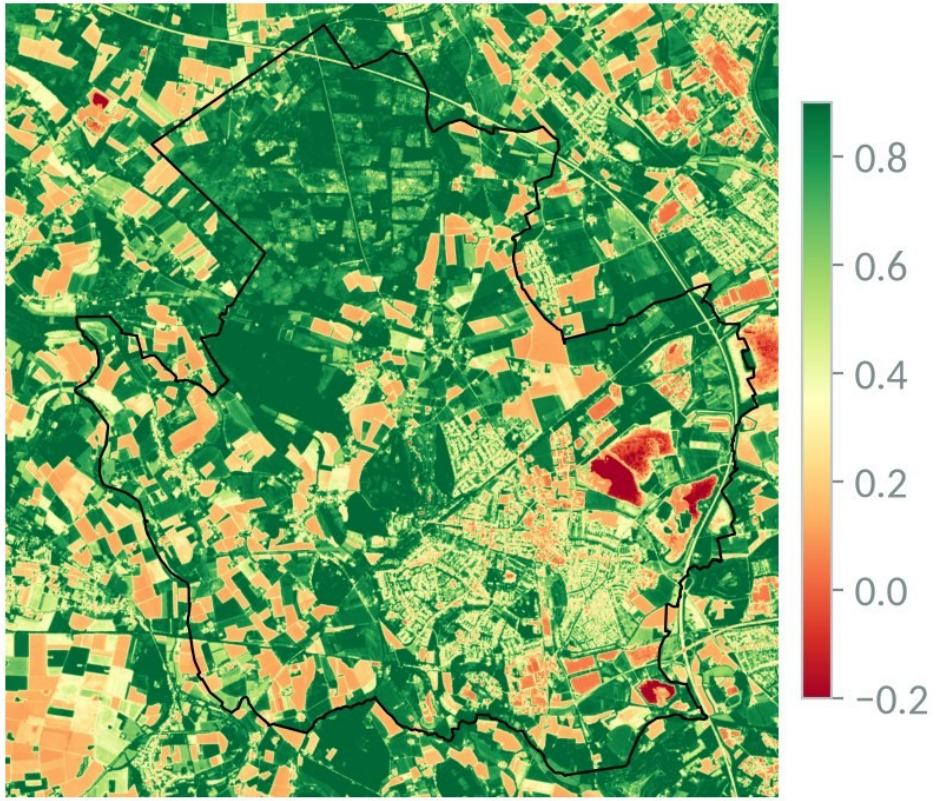
Contains modified Copernicus Sentinel-2 data 2025

From a band stack to an index map

true colour (B04, B03, B02)



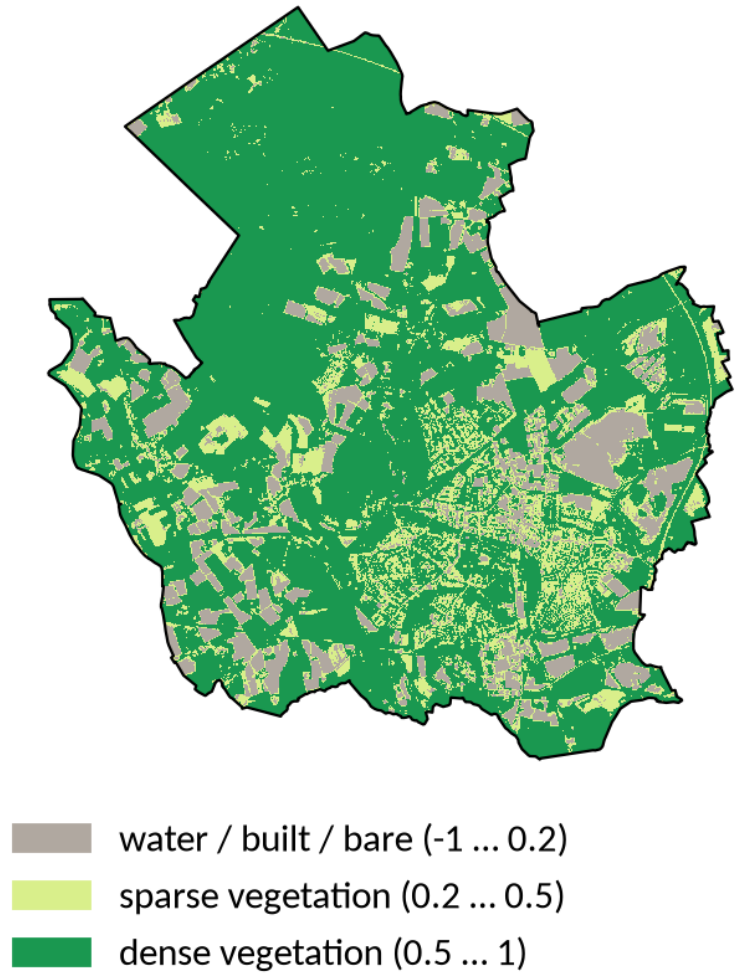
NDVI = (B08 − B04) / (B08 + B04)




```
s2 = rasterio.open("s2_l2a_10m.tif").read()  # B02, B03, B04, B08
```

Contains modified Copernicus Sentinel-2 data 2025

Reclassify: conditions are local operations too



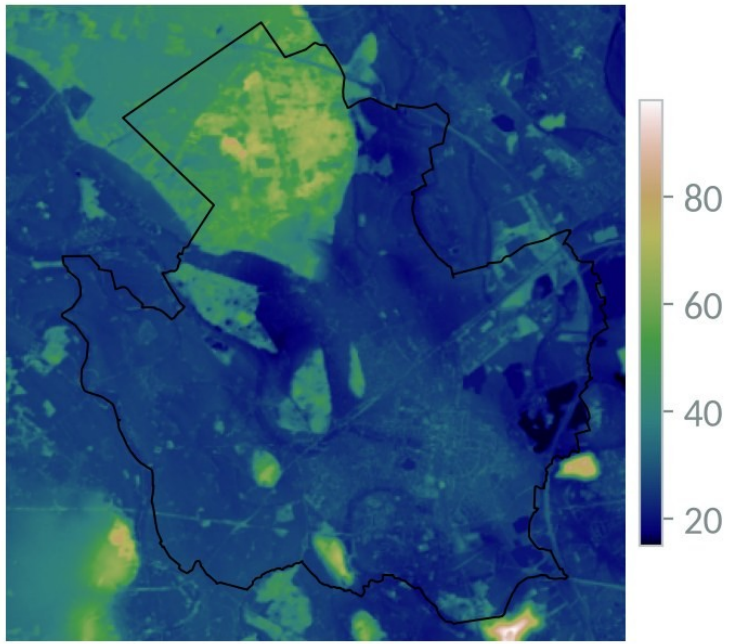
class	area [ha]
water / built / bare	853
sparse vegetation	1 027
dense vegetation	4 435

```
cls = np.select(
    [ndvi < 0.2, ndvi < 0.5],
    [1, 2], default=3)
cls[~inside] = 0   # outside
cells = np.bincount(cls.ravel())
area_ha = cells * 100 / 1e4
```

Contains modified Copernicus Sentinel-2 data 2025

Focal operations: the neighbourhood counts

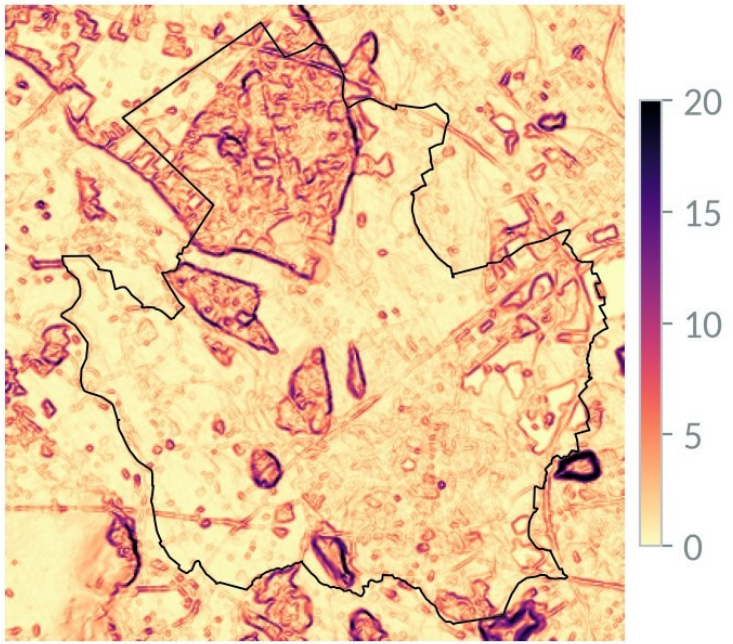
elevation [m] (DEM, 30 m)



hillshade (focal)



slope [°] (focal, 3 × 3)



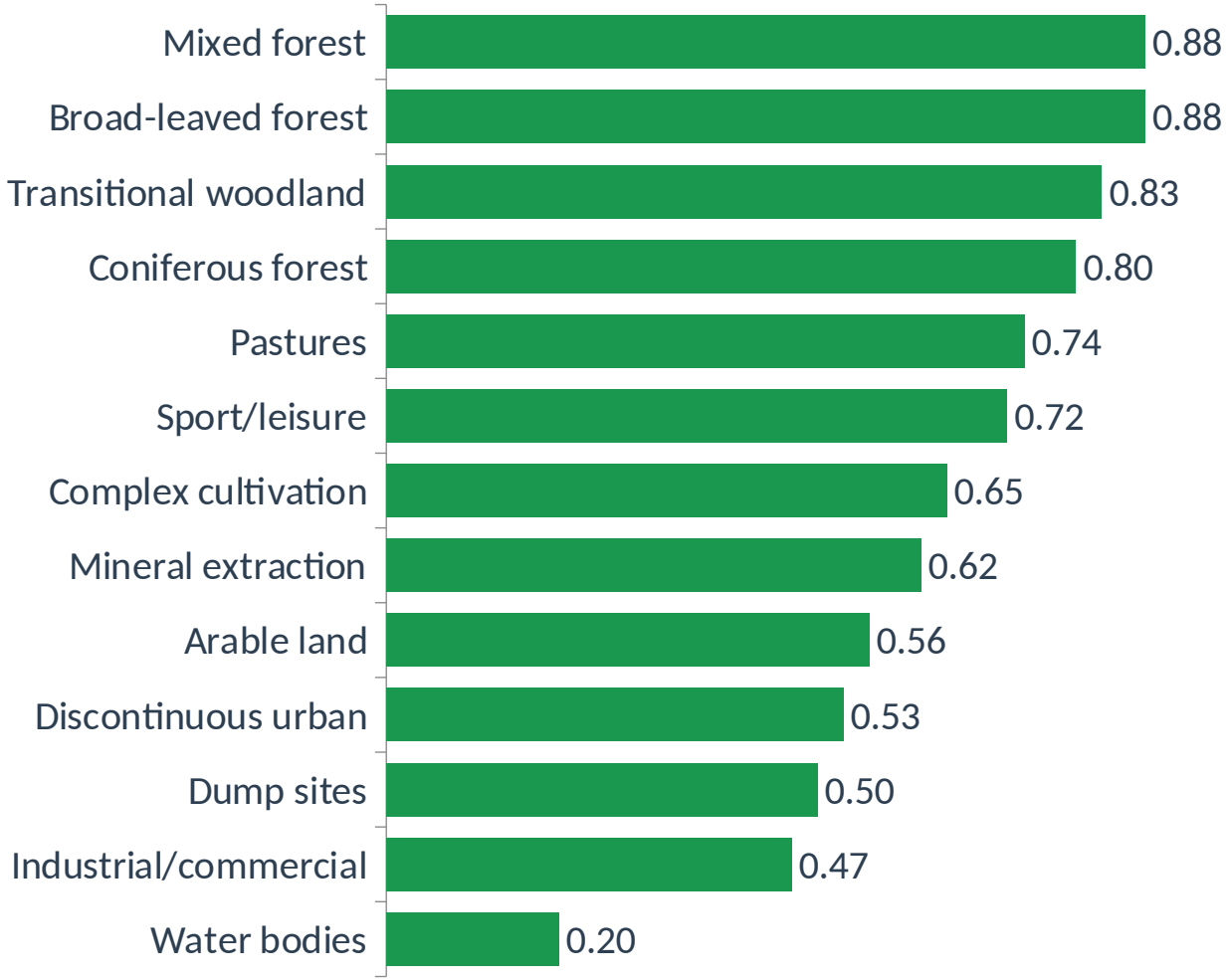

```
gy, gx = np.gradient(dem, 30)
slope = np.degrees(np.arctan(np.hypot(gx, gy)))
smooth = scipy.ndimage.uniform_filter(dem, 3)
```

3 × 3 mean kernel

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Copernicus DEM GLO-30, © DLR e.V. 2010-2014 and © Airbus 2014-2018, ESA

Zonal statistics: mean NDVI per land-cover zone




```
zones = features.rasterize(
  zip(clc.geometry, clc.zone),
  out_shape=ndvi.shape,
  transform=src.transform)
df = pd.DataFrame({"zone":
  zones.ravel(), "ndvi":
  ndvi.ravel()})
df.groupby("zone").ndvi.mean()
```

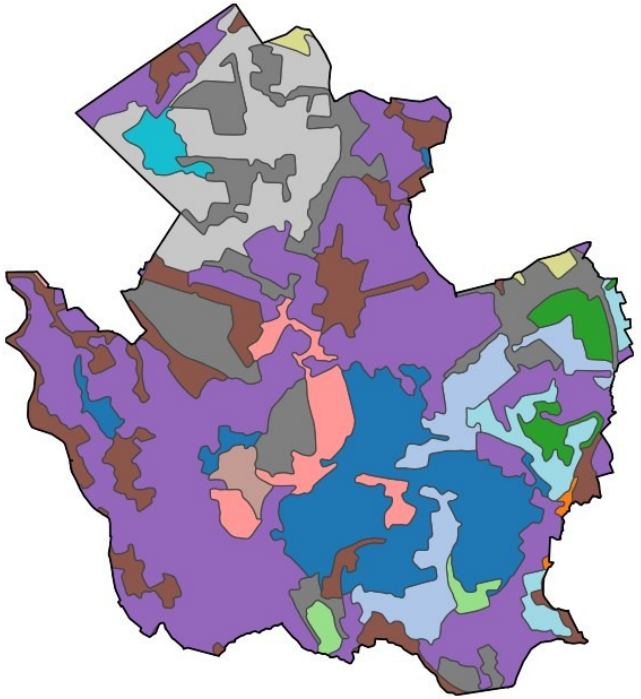
Vector zones are rasterised onto the value grid first.

Contains modified Copernicus Sentinel-2 data 2025 · CORINE Land Cover 2018, © European Union, Copernicus Land Monitoring Service, EEA

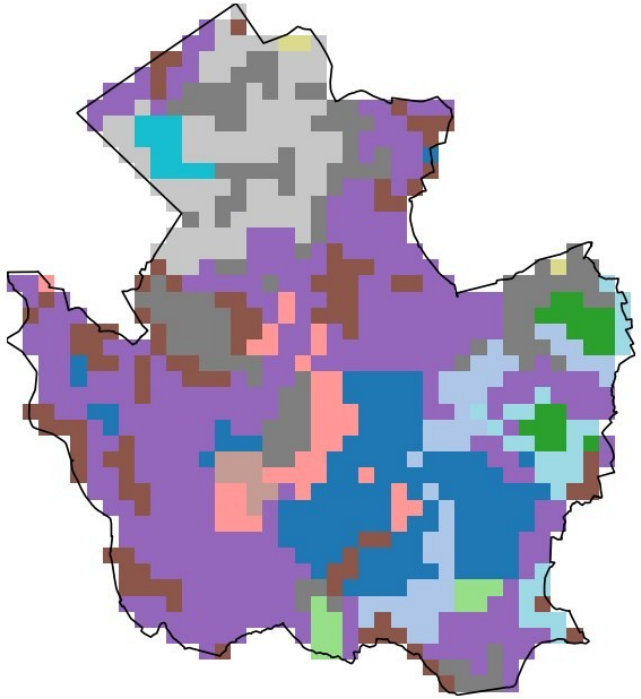


Converting between vector and raster

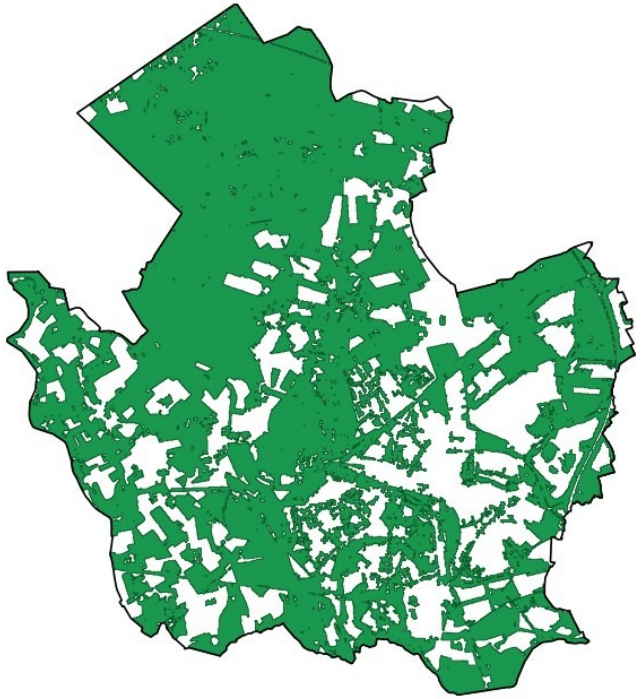
vector: 72 CORINE polygons



rasterize: 250 m cells



polygonize: 43 polygons ≥ 1 ha

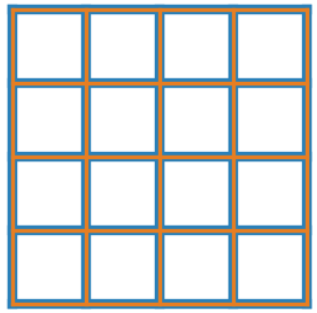


```
features.rasterize(zip(gdf.geometry, gdf.value), out_shape=(h, w), transform=t)
features.shapes(classes, mask=classes == 3, transform=t)  # polygons
```

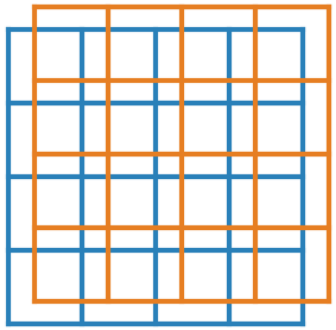
CORINE Land Cover 2018, © European Union, Copernicus Land Monitoring Service, EEA · Contains modified Copernicus Sentinel-2 data 2025

Before combining rasters: align the grids

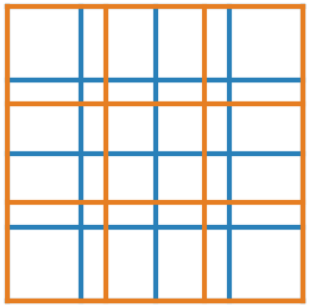
✓ aligned: same CRS, origin and cell size



✗ shifted origin



✗ different cell size



raster A raster B

```
with WarpedVRT(src, crs="EPSG:25832",
        transform=t, width=w, height=h,
        resampling=Resampling.nearest) as v:
    arr = v.read(1)
```

- nearest: classes
- bilinear / cubic: continuous values
- never bilinear for land cover!



Raster operations in three tools

Operation	numpy / rasterio	QGIS	Type
band math, NDVI	(nir - red) / (nir + red)	Raster calculator	local
reclassify	np.select, np.where	Reclassify by table	local
smoothing, slope	scipy.ndimage, np.gradient	Slope, Hillshade	focal
zonal statistics	rasterize + groupby	Zonal statistics	zonal
distance to features	distance_transform_edt	Proximity	global
vector → raster	features.rasterize	Rasterize	conversion
raster → vector	features.shapes	Polygonize	conversion
align / reproject	WarpedVRT, warp.reproject	Warp, Align rasters	preparation



Sources and further reading

- **Data:** Contains modified Copernicus Sentinel-2 data 2025 (S2A_32ULC_20250513, Earth Search STAC); Copernicus DEM GLO-30, © DLR e.V. 2010-2014 and © Airbus 2014-2018, ESA; CORINE Land Cover 2018, © European Union, Copernicus Land Monitoring Service, EEA; municipality boundary © OpenStreetMap contributors (ODbL)
- **Concepts:** C. D. Tomlin (1990): Geographic Information Systems and Cartographic Modeling, Prentice Hall
- **Further reading (links):** rasterio documentation; Earth Lab / ESIIL raster lessons (CC BY-SA); PyGIS raster chapters; QGIS Training Manual – Raster analysis
- **Figures:** own work, generated by make_slides.py in this unit

